

Quick Check

Lesson 5: Trigonometric Ratios of Special Angles (45° , 30° , 60° , 0° , and 90°)

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.4: Trigonometry 1 | Lesson 5

In Construction A, a group draws an isosceles right-angled triangle with both legs (AB and AC) 6 cm long. What is the exact length of the hypotenuse BC?

A $6\sqrt{3}$ cm (about 10.39 cm)

B 12 cm

C $\sqrt{6}$ cm (about 2.45 cm)

D $6\sqrt{2}$ cm (about 8.49 cm)

In Construction B, a group draws an equilateral triangle PQR with side 8 cm and bisects it by drawing PM perpendicular to QR, where M is the midpoint of QR. What is the exact length of PM?

A $4\sqrt{3}$ cm (about 6.93 cm)

B 12 cm

C $4\sqrt{2}$ cm (about 5.66 cm)

D 4 cm

An isosceles right-angled triangle has both legs 1 unit long and a hypotenuse of $\sqrt{2}$ units. What is the exact value of $\sin 45^\circ$?

A $\sqrt{2}$

B 1

C $1/\sqrt{2}$

D $1/2$

A 30-60-90 triangle has sides 1 (opposite 30°), $\sqrt{3}$ (opposite 60°) and 2 (the hypotenuse). What is the exact value of $\tan 30^\circ$?

A $\sqrt{3}/2$

B $1/2$

C $1/\sqrt{3}$

D $\sqrt{3}$

A surveyor in Nakuru stands 40 m from the base of a Kenya Power mast and measures the angle of elevation to the top as exactly 60° . Using $\tan 60^\circ = \sqrt{3}$, what is the height of the mast?

A $40\sqrt{3}$ m (about 69.3 m)

B 80 m

C 40 m

D $40/\sqrt{3}$ m (about 23.1 m)

Why is $\tan 90^\circ$ described as undefined?

- A** $\tan 90^\circ = 1$, because the opposite side and the hypotenuse become equal
- B** $\tan 90^\circ = 0$, because the adjacent side becomes 0
- C** The adjacent side approaches 0, and division by zero is not possible
- D** The opposite side becomes 0, so there is nothing to divide

After today's lesson, a group asks: 'Can we find the exact height of the Nakuru mast if the angle of elevation is 50° ?' Which answer is correct?

- A** No, a mast can only be measured when the angle is 30° , 45° or 60°
- B** Yes, use the 60° triangle, because 50° is close to 60° , and the answer will be exact
- C** The two special triangles give exact values only for 0° , 30° , 45° , 60° and 90° , so for 50° we use tables or a calculator
- D** Yes, use the 45° triangle, because 50° is close to 45° , and the answer will be exact

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).